



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: Jetty-1	AREA: B
PERMIT TO WORK No.: -	TYPE OF SCAFFOLD: Independent Return
SCAFFOLD DRAWING No: PMS-WA-ARCO-039	
RISK CLASSIFICATION: ☒ High ☐ Medium ☐ Low	

SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an Independent Return scaffold required for maintenance of the condensate loading arm, equipment tag Z-3301A,Z-3301B,Z-3301C at Jetty-1. The scaffold envelope is 5m x 4m x 20m (length x width x height) and is configured with standards, ledgers, transoms, bracing, platforms, and support/tie arrangements as indicated on the approved drawing.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Total Horizontal Load in the X Direction (F_X): 2.103 kN

Total Horizontal Load in the Z Direction (F_Z): 4.807 kN

Role	ID	Name	Signature	Date
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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	Independent Return	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-038	Prepared By: Nnamani, Ifeanyi	

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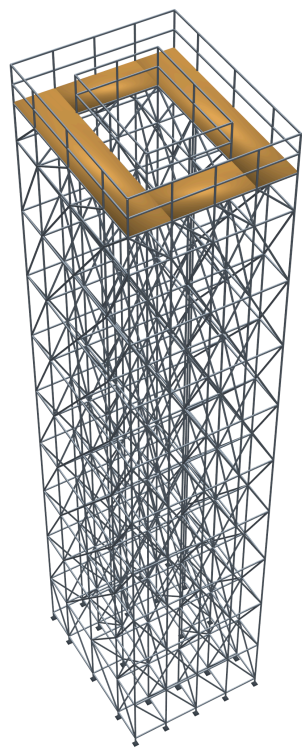
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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 / 10.3.2.2

Scaffold tube material has a minimum nominal yield strength of 235 N/mm² (BS EN 12811-1, 4.2.1.2). A partial safety factor $\gamma_M = 1.1$ (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm² used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 10.2.3.2 / 10.2.3.3

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 17.522$ kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot.

LOAD CASES

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1

Platform live loading is applied to the scaffold working platforms as member UDLs in the STAAD model. The load intensity is multiplied by the tributary width of each supporting member to obtain the applied member line load. (LC 2).

Member	Tributary Width (m)	Load Intensity (kN/m ²)	Member Load (kN/m)
Members 661, 664, 667, 671, 681, 682, 686, 692 (Edge)	$1.000 / 2 = 0.500$	1.500	0.750
Members 669, 684 (Interior)	$(0.500 + 0.500) = 1.000$	1.500	1.500
Members 666, 672 (Interior)	$(0.500 + 0.750) = 1.250$	1.500	1.875
Members 678, 688 (Interior)	$(0.750 + 0.500) = 1.250$	1.500	1.875
Members 675, 690 (Interior)	$(0.750 + 0.750) = 1.500$	1.500	2.250

Load Intensity = 1.50 kN/m² — BS EN 12811-1 Class 2

Member Load = Load Intensity × Tributary Width. Edge members carry half the adjacent bay width; interior members carry half the bay from each side.
 Governing load intensity (SWL) = **1.5 kN/m²** — BS EN 12811-1 Class 2.
 Total applied platform live load = **21.0 kN**.

Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **3.6 kN** (member UDL 0.15 kN/m)
Handrail Load (HLZ) = **4.8 kN** (member UDL 0.15 kN/m)

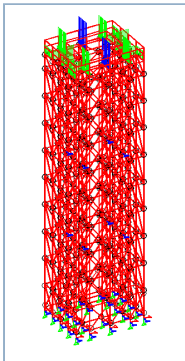
Wind Load

EN 1991-1-4:2005

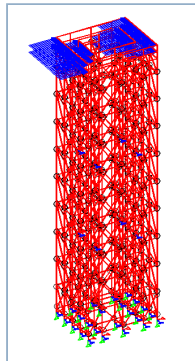
Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.077357 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load - X Axis (WLX) = **54.05 kN**
Total Wind Load - Z Axis (WLZ) = **58.06 kN**

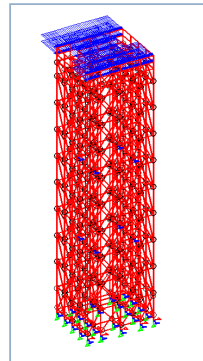
Load Diagrams



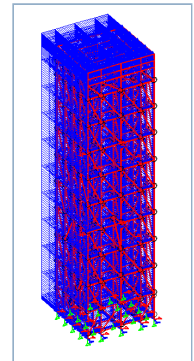
Live Load (LL)



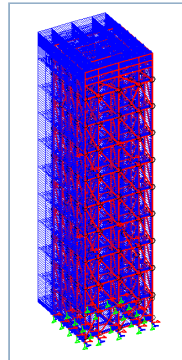
Handrail Load X (HLX)



Handrail Load Z (HLZ)



Wind Load X (WLX)



Wind Load Z (WLZ)

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Durst, C.S. (1960)	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	25.11	m/s
	Basic wind speed $V_b = V600 =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	Z _{min}	1.0	m

Reference	Calculation	Value	Unit
	z - height of structure under consideration	20.0	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ - Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_0,ii]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
Wind Turbulence			
	$C_r(z) = K_r \times \ln(z/z_0)$	1.3739	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	34.492	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity $I_v(z) = k1 / [C_o(z) \times \ln(z/z_0)]$	0.1136	
Peak Velocity Pressure			
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \times I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	1334.6649	N/m²
Wind Load = $C_f \times q_p \times A_f$			
$C_f =$		1.2	
$q_p =$		1.334665	kN/m²
$A_f =$	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.077357	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**;
SLS (characteristic) combinations use gamma = **1.0**.
DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	SLS DL + LL	LC1 x 1.0 + LC2 x 1.0
8	SLS DL + LL + HLX	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
9	SLS DL + LL + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
10	SLS DL + WLX	LC1 x 1.0 + LC5 x 1.0
11	SLS DL + WLZ	LC1 x 1.0 + LC6 x 1.0
12	SLS DL + LL + WLX	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
13	SLS DL + LL + WLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
14	SLS DL + LL + WLZ + HLX	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC3 x 1.0
15	SLS DL + LL + WLZ + HLZ	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0 + LC4 x 1.0
16	ULS 1.5DL + 1.5LL	LC1 x 1.5 + LC2 x 1.5
17	ULS 1.5DL + 1.5LL + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
18	ULS 1.5DL + 1.5LL + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
19	ULS 1.5DL + 1.5LL + 1.5WLX	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
20	ULS 1.5DL + 1.5LL + 1.5WLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
21	ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLX	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC3 x 1.5
22	ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLZ	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5 + LC4 x 1.5
23	ULS 1.5DL + 1.5WLX	LC1 x 1.5 + LC5 x 1.5
24	ULS 1.5DL + 1.5WLZ	LC1 x 1.5 + LC6 x 1.5
25	ULS DL + 1.5WLX	LC1 x 1.0 + LC5 x 1.5

LC No.	Combination	Component Loadings & Factors
26	ULS DL + 1.5WLZ	LC1 x 1.0 + LC6 x 1.5

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.352 at Member No. **1500**, Load Combination **22**, Clause **EC-6.2.9.1**. **PASS** - All members adequate; UC ratio below unity.

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
1500	PIP483.6	PASS	EC-6.2.9.1	0.352	22	0.26	C
1310	PIP483.6	PASS	EC-6.3.3-661	0.312	22	11.97	C
1482	PIP483.6	PASS	EC-6.2.9.1	0.312	22	0.19	C
477	PIP483.6	PASS	EC-6.3.3-662	0.310	22	13.40	C
690	PIP483.6	PASS	EC-6.2.9.1	0.304	22	0.91	T
1304	PIP483.6	PASS	EC-6.3.3-661	0.304	22	11.66	C
1497	PIP483.6	PASS	EC-6.2.9.1	0.301	22	0.20	C
462	PIP483.6	PASS	EC-6.3.3-662	0.300	22	12.91	C
1487	PIP483.6	PASS	EC-6.2.9.1	0.284	22	0.27	C
688	PIP483.6	PASS	EC-6.3.3-662	0.265	22	0.28	C
1309	PIP483.6	PASS	EC-6.3.3-661	0.265	22	10.08	C
672	PIP483.6	PASS	EC-6.3.3-662	0.255	22	1.73	C
678	PIP483.6	PASS	EC-6.3.3-662	0.251	22	1.80	C
666	PIP483.6	PASS	EC-6.2.9.1	0.247	22	0.03	C
675	PIP483.6	PASS	EC-6.2.9.1	0.247	22	0.21	T
1490	PIP483.6	PASS	EC-6.2.9.1	0.245	22	0.21	C
478	PIP483.6	PASS	EC-6.3.3-662	0.244	22	10.04	C
463	PIP483.6	PASS	EC-6.3.3-662	0.240	22	9.82	C
647	PIP483.6	PASS	EC-6.3.3-662	0.240	22	0.89	C
1486	PIP483.6	PASS	EC-6.2.9.1	0.240	22	0.22	C
473	PIP483.6	PASS	EC-6.3.3-662	0.223	22	8.91	C
1327	PIP483.6	PASS	EC-6.3.3-661	0.223	19	6.65	C
464	PIP483.6	PASS	EC-6.3.3-662	0.221	22	8.82	C
1303	PIP483.6	PASS	EC-6.3.3-661	0.220	22	8.32	C
1201	PIP483.6	PASS	EC-6.3.3-661	0.216	21	8.16	C
1219	PIP483.6	PASS	EC-6.3.3-661	0.215	19	6.41	C
442	PIP483.6	PASS	EC-6.2.9.1	0.207	22	0.17	T
480	PIP483.6	PASS	EC-6.3.3-662	0.206	22	8.84	C
1273	PIP483.6	PASS	EC-6.3.3-661	0.206	22	7.77	C
427	PIP483.6	PASS	EC-6.2.9.1	0.203	22	0.43	T
431	PIP483.6	PASS	EC-6.3.3-662	0.200	22	0.17	C
650	PIP483.6	PASS	EC-6.3.3-662	0.200	22	1.28	C
1213	PIP483.6	PASS	EC-6.3.3-661	0.199	19	5.86	C
1328	PIP483.6	PASS	EC-6.3.3-661	0.197	19	7.41	C
1489	PIP483.6	PASS	EC-6.2.9.1	0.197	22	0.25	T
420	PIP483.6	PASS	EC-6.3.3-662	0.196	22	8.34	C
1196	PIP483.6	PASS	EC-6.3.3-661	0.196	22	7.35	C
1485	PIP483.6	PASS	EC-6.2.9.1	0.196	22	0.24	T
452	PIP483.6	PASS	EC-6.2.9.1	0.195	22	1.94	T
646	PIP483.6	PASS	EC-6.3.3-662	0.195	22	1.10	C

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
1212	PIP483.6	PASS	EC-6.3.3-661	0.195	19	5.74	C
416	PIP483.6	PASS	EC-6.3.3-662	0.192	22	8.19	C
1195	PIP483.6	PASS	EC-6.3.3-661	0.190	21	7.12	C
1194	PIP483.6	PASS	EC-6.3.3-661	0.189	20	7.06	C
475	PIP483.6	PASS	EC-6.3.3-662	0.188	19	7.66	C
472	PIP483.6	PASS	EC-6.3.3-662	0.187	19	7.56	C
1321	PIP483.6	PASS	EC-6.3.3-661	0.186	19	5.46	C
1494	PIP483.6	PASS	EC-6.2.9.1	0.186	21	0.22	C
297	PIP483.6	PASS	EC-6.3.3-662	0.185	22	7.69	C
537	PIP483.6	PASS	EC-6.3.3-662	0.185	22	7.57	C
Showing highest 50 utilization ratios; 1021 lower-utilization members omitted.							

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **3.740 kN** (Member 428, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 3.740 kN ≤ 10.0 kN → **PASS**

Class A right-angle couplers are adequate.

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force F _{s,k} (kN)	10.0	15.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.8
	Pull-apart force F _{p,k} (kN)	20.0	30.0
	Rotational moment M _{T,k} (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force F _{s,k} (kN)	6.0	9.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.8
	Pull-apart force F _{p,k} (kN)	15.0	22.5
	Rotational moment M _{T,k} (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force F _{s,k} (kN)	-	3.0
	Cruciform bending moment M _{B,k} (kNm)	-	0.6

Member Axial Force Summary (Top 30 Horizontal Members — ULS Code Check)

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
428	C	PASS	EC-6.3.3-662	0.117	22	3.740
503	C	PASS	EC-6.3.3-662	0.097	22	3.520
683	C	PASS	EC-6.3.3-662	0.076	22	3.250
1471	C	PASS	EC-6.3.3-661	0.069	22	3.120
490	C	PASS	EC-6.3.3-662	0.086	22	3.080
563	C	PASS	EC-6.3.3-662	0.061	22	3.020
550	C	PASS	EC-6.3.3-662	0.063	22	2.710
548	C	PASS	EC-6.3.3-662	0.058	22	2.680
668	C	PASS	EC-6.3.3-662	0.118	22	2.570

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
629	T	PASS	EC-6.2.9.1	0.100	22	2.410
608	C	PASS	EC-6.3.3-662	0.056	22	2.340
696	C	PASS	EC-6.3.3-662	0.103	22	2.240
569	T	PASS	EC-6.2.9.1	0.114	22	2.230
573	C	PASS	EC-6.3.3-662	0.058	22	2.110
693	C	PASS	EC-6.3.3-662	0.111	22	2.060
451	T	PASS	EC-6.2.9.1	0.160	22	2.030
633	C	PASS	EC-6.3.3-662	0.065	22	1.950
452	T	PASS	EC-6.2.9.1	0.195	22	1.940
678	C	PASS	EC-6.3.3-662	0.251	22	1.800
509	T	PASS	EC-6.2.9.1	0.134	22	1.760
677	C	PASS	EC-6.3.3-662	0.136	22	1.750
672	C	PASS	EC-6.3.3-662	0.255	22	1.730
512	T	PASS	EC-6.2.9.1	0.125	22	1.700
623	C	PASS	EC-6.3.3-661	0.045	22	1.700
1474	C	PASS	EC-6.3.3-662	0.044	21	1.680
391	T	PASS	EC-6.2.9.1	0.131	22	1.630
607	T	PASS	EC-6.2.9.1	0.072	26	1.610
425	T	PASS	EC-6.2.9.1	0.168	22	1.600
496	T	PASS	EC-6.2.9.1	0.129	22	1.580
493	T	PASS	EC-6.2.9.1	0.119	22	1.530

FRICTIONAL RESISTANCE

Max vertical reaction per support ($\text{Max } f_y$) = Resistance x 0.15 / C_F = **8.761 kN**

Coefficient of friction (C_F) = **0.3**

Frictional Resistance (FR) = $C_F \times \text{Max } f_y$ = **2.6283 kN**

Spring Stiffness = FR / 0.15 = **17.522 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results.
Displacement tables are shown below.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 15 Member 1557 Member length L = 1500.0 mm	0.798 mm	$L/100 = 1500/100 = 15.0 \text{ mm}$	PASS

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
1557	0.790	0.750	15	1897	0.120	0.074	9.322
1560	0.787	0.750	15	1906	0.119	0.074	9.403
675	0.657	0.417	15	1522	0.132	-0.688	4.444
1571	0.630	0.750	15	2382	-0.214	-0.851	7.999
1568	0.608	0.750	15	2468	-0.214	-0.859	7.983
675	0.578	0.417	9	1728	-0.040	-0.856	1.110
675	0.548	0.500	13	1824	0.070	-0.710	3.284
675	0.546	0.500	14	1829	0.389	-0.707	3.332
1509	0.527	0.875	15	2845	0.121	0.078	6.486
1560	0.518	0.750	9	2893	-0.096	-0.248	4.614
1512	0.518	0.875	15	2896	0.120	0.079	6.533
1557	0.514	0.750	9	2918	-0.095	-0.257	4.552
678	0.511	0.583	15	1957	0.175	-0.540	2.721
672	0.504	0.583	15	1982	0.083	-0.529	2.621

Max = **0.798 mm** (LC 15) | L = Member 1557 length (1500 mm) | Allowable L/100 = 15.0 mm → **PASS**

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 15 Member 647 Member length L = 1000.0 mm	0.425 mm	L/200 = 1000/200 = 5.0 mm	PASS

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
647	0.425	0.667	15	2352	0.125	0.110	3.898
647	0.342	0.667	9	2921	-0.090	-0.217	0.714
650	0.337	0.667	15	2970	0.121	0.090	2.288
646	0.332	0.667	15	3013	0.121	0.090	2.195
587	0.303	1.167	15	6600	0.136	0.118	4.108
590	0.275	1.167	15	7279	0.139	0.107	2.401
586	0.273	1.167	15	7314	0.130	0.104	2.320
1500	0.268	0.208	15	1868	-0.239	-0.828	4.462
647	0.260	0.583	14	3848	0.313	-0.040	2.980
647	0.259	0.583	13	3855	0.029	-0.052	2.939
587	0.246	1.167	13	8142	0.049	-0.036	3.162
587	0.243	1.167	14	8215	0.310	-0.027	3.182
590	0.236	1.167	14	8489	0.314	-0.059	1.895
650	0.235	0.667	14	4264	0.318	-0.078	1.789

Max = **0.425 mm** (LC 15) | L = Member 647 length (1000 mm) | Allowable L/200 = 5.0 mm → **PASS**

CONCLUSION

The structural analysis and design of the **Independent Return** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.352** at Member 1500). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 15.0mm** and **L/200 = 5.0mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and

material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

INSTALLATION NOTES

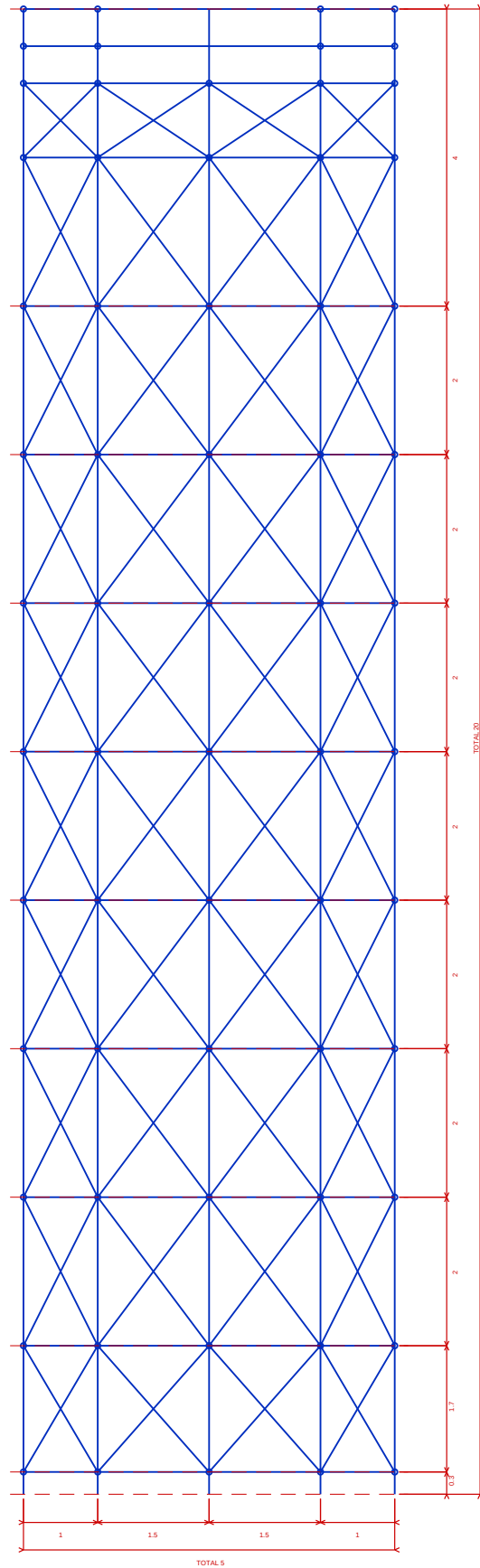
1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Independent Return scaffold configuration matches the approved drawing and is suitable for maintenance.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Set out and level the scaffold base plates and sole boards in accordance with the approved working drawing.
8. Confirm the scaffold footprint/envelope is approximately 5.0m x 4.0m x 20.0m or as otherwise dimensioned on the approved drawing.
9. Erect scaffold tube standards (48.3 x 3.6 mm), ledgers, and transoms progressively from the base, checking plumb and level in both axes before proceeding to the next lift.
10. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to form the first stable frame.
11. Install mid-level horizontal ledgers and transoms at 2.0m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 4.0m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 6.0m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 8.0m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 10.0m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 12.0m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 14.0m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 16.0m height in accordance with the working drawing.
19. Install mid-level horizontal ledgers and transoms at 18.0m height in accordance with the working drawing.
20. Install mid-level horizontal ledgers and transoms at 19.0m height in accordance with the working drawing.
21. Install mid-level horizontal ledgers and transoms at 19.5m height in accordance with the working drawing.
22. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
23. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
24. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
25. Gravlock or box-tie the scaffold structure to the existing structure at all indicated tie points at 6.0m and 12.0m height as shown on the working drawing.
26. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
27. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
28. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer.
End-user or third-party modification of this scaffold structure is strictly prohibited.
29. For dismantling, perform the erection steps in reverse order.

REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design — Load Combinations, Partial Factors, and Limit States (ULS/SLS/EQU)	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Durst, C.S.	Wind Speeds Over Short Periods of Time. The Meteorological Magazine, 89, 181–186 — basis of 3-second gust to 10-minute mean conversion	1960
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

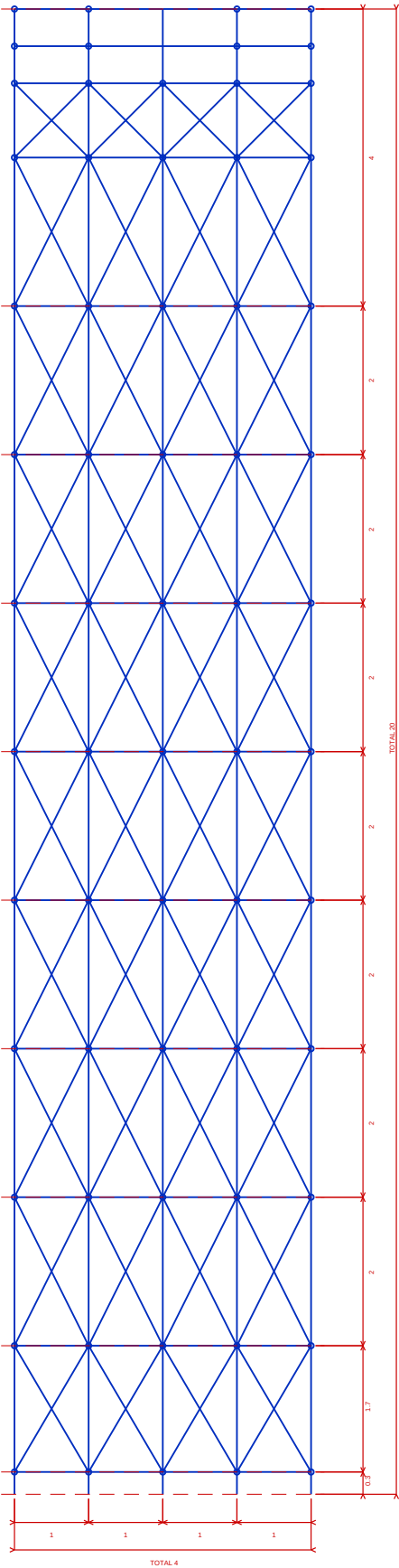
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



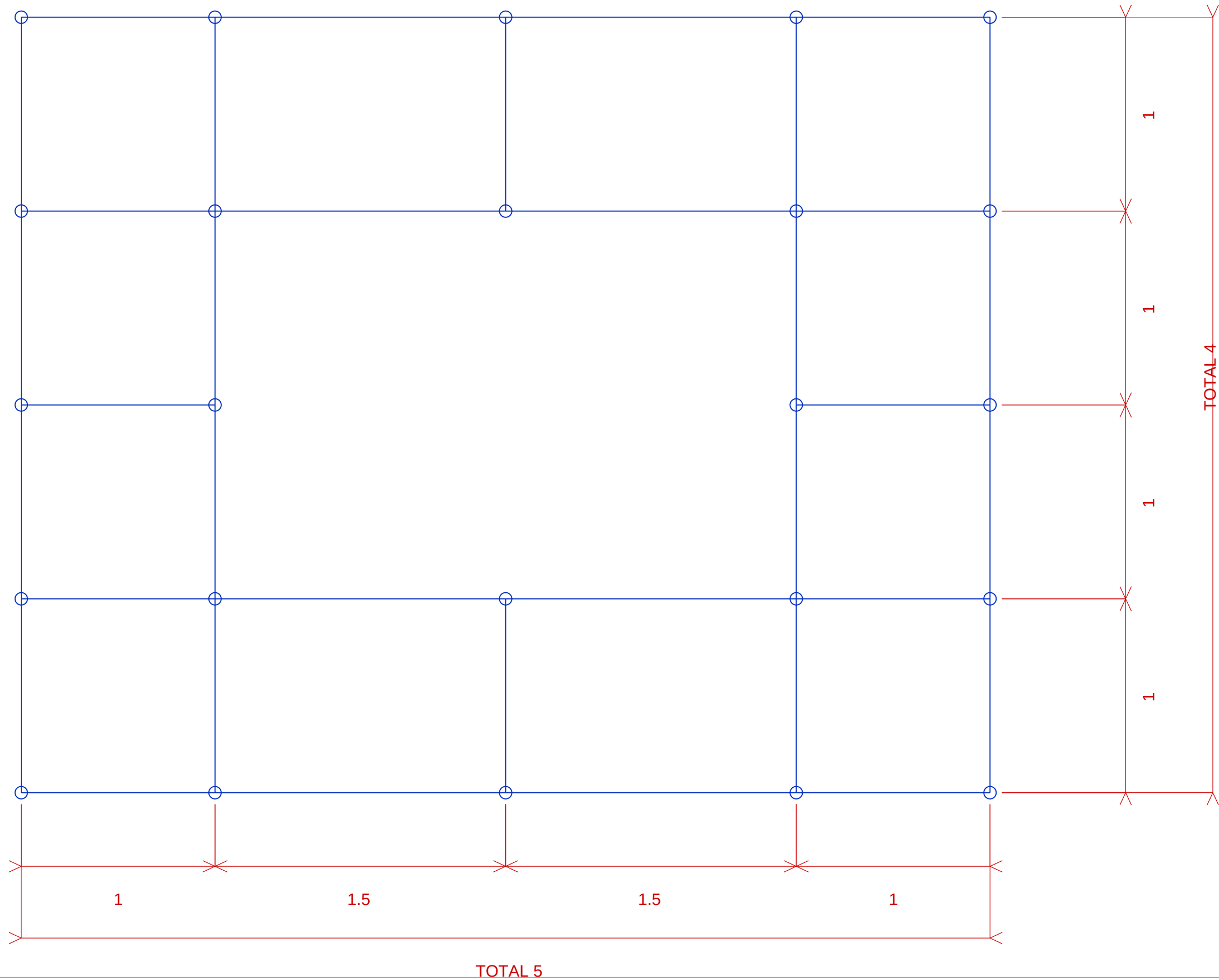
FRONT VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-038-FRONT-VIEW			 	
DRAWING TITLE PMS-WA-ARCO-038 - FRONT VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0	COMPANY	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	SHEET 1 OF 4	SIZE A3



SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-038-SIDE-VIEW			 
DRAWING TITLE PMS-WA-ARCO-038 - SIDE VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY
					SHEET 2 OF 4
					SIZE A3

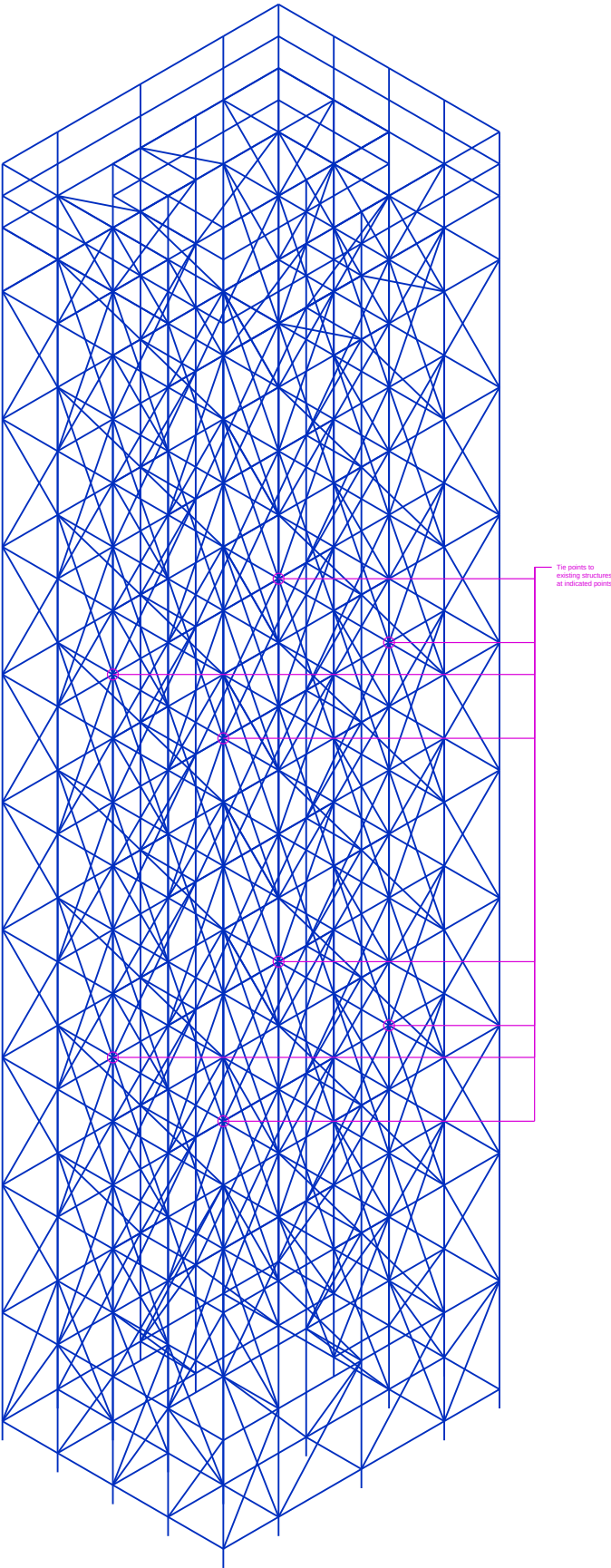


TOTAL 5

TOTAL 4

PLAN VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-038-PLAN-VIEW			 
DRAWING TITLE PMS-WA-ARCO-038 - PLAN VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY
					SHEET 3 OF 4
					SIZE A3



The points to
existing structures
at indicated points.

3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-038-3D-VIEW			 
DRAWING TITLE PMS-WA-ARCO-038 - 3D VIEW		SCALE NTS	DATE 07-JUL-2026	REV 0	
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	REVIEWED Ezewoko, Joseph	APPROVED Chima, Tasieobi	COMPANY
					SHEET 4 OF 4
					SIZE A3